



InfoVerse@AU

Sprint: Programming the Web

Presented By :

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PROJECT SCOPE

Team

3–4 members collaborating on design, coding, and analysis

- Varshan – UI/UX Designer & HTML,CSS,JS
- Manoj – API Integration & HTML,CSS,JS
- Yeshwanth – Deployment Lead & HTML,CSS,JS

Goals

- ✓ Build an interactive webpage using two public APIs
- ✓ Implement DOM manipulation and event listeners
- ✓ Analyze browser–server communication using Wireshark

Tools & Technologies

- HTML, CSS, JavaScript (Fetch API, Async/Await)
- Wireshark for network packets tracing
- GitHub Pages / Netlify / Vercel for deployment

Project Expectations

Project Deliverables

Deliverables

- We will develop an interactive webpage using HTML, CSS, and JavaScript.
- Our page will use two public APIs –AdviceSlip API and Currency API to display live data.
- We will build and test the project in VS Code using the Live Server extension.
- After development, we will host the webpage online using GitHub Pages or Vercel.
- We will capture and analyze API communication using Wireshark.
- Our final deliverables will include the source code, screenshots, Wireshark captures, an analysis report, and the hosted webpage link.

Constraints

- We will use only free and open APIs for our data sources.
- The entire project will be done using HTML, CSS, and JavaScript only no external frameworks.
- Our webpage will show real-time DOM updates and interactivity through JavaScript.
- The hosted version will run directly in a browser without any additional setup or dependencies.
- We will design the site to be responsive and user-friendly on all screen sizes.

PROJECT SCOPE & DELIVERABLES

Phase



Phase 1: Design & Setup

Phase 2: Development & Integration

Phase 3: Testing, Analysis & Hosting

Scope



- **Purpose:** Digital college newsletter with live updates.
- **APIs:**
 - **AdviceSlip:** Gives advice
 - **Currency API** – live exchange rates
- **Layout:** Header with logo/time, currency section, news feed.
- **Goal:** Quick, simple, and responsive info hub.

- **Fetch:** Use *fetch()* to get JSON from APIs.
- **Dynamic Elements:** *#advice*, *#currency-section*.
- **Interactivity:** Buttons refresh data; *setInterval()* updates time.

- **Capture:** Use Wireshark to monitor API traffic.
- **Observe:** HTTP method (GET/POST), status, latency, payload.
- **Hosting:** GitHub Pages or Vercel.

Deliverables



- We will start by designing the basic layout and wireframe for our webpage.
 - We plan to create the HTML, CSS, and JavaScript project structure in VS Code.
 - We have finalized two APIs for our project – AdviceSlip API, Currency API
 - We will set up Live Server in VS Code to test our webpage during development.

- We will integrate the AdviceSlip API, Currency, using the Fetch API.
 - We will use Async/Await for smooth and efficient API data loading.
 - We plan to add event listeners and dynamic DOM updates for interactivity using JavaScript.

- We will use Wireshark to capture and analyze API communication.
 - We will host our webpage using GitHub Pages or Vercel.
 - Finally, we will prepare a 2–3 page analysis report with screenshots and observations.

● APIs Used

APIs Integrated & Purpose

- **Frankfurter Currency API –**

Fetches real-time currency exchange rates (e.g., [https://api.frankfurter.app/latest?from=\\${base}&to=INR](https://api.frankfurter.app/latest?from=${base}&to=INR)) allows users to view live currency conversions and stay updated with current rates.

- **AdviceSlip API –**

Fetches motivational quotes and advice (e.g., <https://api.adviceslip.com/advice>) keeps users inspired and engaged with a new quote every refresh or interval.